Technical Reproducibility Validation Report

GangSTR

Version v2.5
Autosomal STR

Verification date	2026-11-01
Repository commit	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Result	**Runs from its published image; the pinned commit does not build**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/gangstr
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs from its published image; the pinned commit does not build**
REACHED	Runs + Plausible output, 5/5 gates passed
WHY	The tool's run produced its documented output, on the published environment the README points at: the build from the pinned commit failed, so what ran is the version that environment holds.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	gangstr.log-external.txt	

3. Not Built From Source: the README's Ready-Made Environment Ran

STRhub tried to build the tool from its source at the pinned commit, following the build steps the repository declares, and the build failed. The published image gymreklab/str-toolkit the README points at was used instead, and every gate below ran on it.

What this means

– **If you are trying to run it:** A build from source at this commit fails in a clean environment; the cause is below. The ready-made environment the README points at does work: it is what this run used.

– **If you are reviewing a paper:** This result describes the software inside that environment, whatever its publisher last put there, and not the pinned commit, which is the version a manuscript would cite.

– **If you maintain it:** Every cause identified sits in this run's configuration: the pinned versions, package names or build steps it declared. They are faults in how the tool was set up here rather than in the software. Re-verifying after correcting them is free.

What failed

What happened	Suggested fix

An autotools build is missing its auxiliary file: config.sub	The configure script needs 'config.sub', which `autoreconf -i` (or `automake --add-missing`) copies in from the automake package. The build runs autoconf without installing those files; running `autoreconf -fi` before `./configure` in that step fixes it.

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · GangSTR v2.5 · commit 6ea9b2b
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( GangSTR \
--bam /data/in/input.bam \
--ref /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--out outprefix \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

5. Reproducibility Metadata

Tool	GangSTR
Version	v2.5
Repository	https://github.com/gymreklab/gangstr
Commit (pinned)	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-11-01 12:50:13 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/41308001752

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	outprefix.vcf
Format	VCF
Sequence records	24
Distinct STR loci	24
Total reads	11,603
Max depth (single locus)	800

8. Verification Data

The repository ships example data (10 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)

Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA
Regions BED	Provided by STRhub

9. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

– A container environment: the build from the pinned commit failed, so the published image gymreklab/str-toolkit the README points at was built instead. What ran is the version that environment holds, not necessarily the pinned commit.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's CMakeLists.txt (cmake), at the pinned commit. If that build fails, the published image gymreklab/str-toolkit the README points at stands in, and the report says so.
– Input: the README documents BAM (line 6); this run used STRhub's illumina-bam-hg38 reference data.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– 1 BED path(s) replaced with /data/in/regions.bed.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.
– Regions: STRhub's ready-made gangstr file for the dataset's panel loci (STRhub records the gangstr layout for a program of this name).

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

◆ **Runs end-to-end**

GangSTR v2.5 installs and executes without error in an environment built from the published image gymreklab/str-toolkit the README points at, after the build from the pinned commit failed.

◆ **Produces plausible output**

The tool generated a structurally valid VCF output file with genotype calls across 24 target forensic STR loci, with a maximum read depth of 800 at D18S51.

◆ **Verdict: Runs**

The tool's run produced its documented output, on the published environment the README points at: the build from the pinned commit failed, so what ran is the version that environment holds.

◆ **Repository test data not used**

The repository ships example data (10 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
D18S51	800	■■■■■■■■■■■■■■■■■■	**D13S317**	476	■■■■■■■■■
D7S820	684	■■■■■■■■■■■■■	**PentaE**	473	■■■■■■■■■
D16S539	656	■■■■■■■■■■■■■	**CSF1PO**	447	■■■■■■■■
D5S818	641	■■■■■■■■■■■■	**D19S433**	394	■■■■■■■
PentaD	621	■■■■■■■■■■■■	**D12S391**	379	■■■■■■■
D6S1043	609	■■■■■■■■■■■■	**SE33**	343	■■■■■■
D3S1358	593	■■■■■■■■■■■	**TPOX**	340	■■■■■■
D2S441	582	■■■■■■■■■■■	**TH01**	322	■■■■■■
D8S1179	578	■■■■■■■■■■■	**D21S11**	303	■■■■■■
vWA	568	■■■■■■■■■■■	**D1S1656**	268	■■■■■
D22S1045	529	■■■■■■■■■■	**D2S1338**	259	■■■■■
D10S1248	484	■■■■■■■■■	**FGA**	254	■■■■■